

Introduction Manual

Canvas Assignment AI Evaluator System

Beginner-Friendly Guide for Faculty and Non-Technical Users

1. Introduction

The **Canvas Assignment AI Evaluator System** is an intelligent automation platform designed to simplify, accelerate, and modernize the process of academic assignment evaluation.

The system integrates:

- Canvas Learning Management System (LMS)
- Artificial Intelligence evaluation
- Automated grading workflows
- Rubric-based assessment
- Automatic report generation

into one complete solution.

In traditional academic environments, assignment evaluation is one of the most time-consuming tasks for faculty members. Teachers often spend several hours or even days downloading submissions, organizing files, checking answers, assigning marks, writing comments, preparing reports, and uploading grades back into the LMS.

This project was created to solve those problems using automation and AI.

The system automatically:

1. Connects to Canvas
2. Downloads all student submissions
3. Reads student files
4. Evaluates assignments using AI
5. Generates marks and feedback
6. Uploads grades back to Canvas
7. Creates downloadable reports

The goal is not to replace teachers, but to reduce repetitive manual work and assist faculty members in handling large academic workloads more efficiently.

2. Why This System Was Needed

Modern educational institutions increasingly rely on online learning platforms such as Canvas LMS. While these systems simplify assignment collection, the actual evaluation process still remains heavily manual.

Faculty members typically face several challenges:

- Large class sizes
- Increasing assignment frequency
- Time pressure
- Administrative workload
- Repetitive grading tasks
- Manual feedback generation
- Grade upload management

For example, consider a faculty member teaching:

- 200 students
- 3 assignments per week
- Multiple file submissions
- Theory and programming assignments

Manually evaluating all submissions may require:

- Downloading hundreds of files
- Organizing folders manually
- Opening every document individually
- Reading submissions one by one
- Writing repetitive feedback
- Uploading grades manually
- Maintaining spreadsheets

This process becomes exhausting, inefficient, and error-prone.

The Canvas Assignment AI Evaluator System was developed specifically to eliminate these repetitive steps.

3. Problems in Traditional Assignment Evaluation

Before understanding the advantages of this system, it is important to understand the limitations of traditional evaluation methods.

3.1 Manual Downloading

Normally, faculty members:

- Open Canvas
- Download submissions manually
- Save files individually
- Rename files
- Organize folders manually

This becomes extremely difficult for large classes.

3.2 File Management Problems

Students upload different types of files:

- PDFs
- DOCX documents
- ZIP files
- Code files
- Images
- Text submissions

Managing these manually becomes chaotic.

Common problems include:

- Duplicate files
 - Incorrect filenames
 - Missing submissions
 - Disorganized folders
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3.3 Repetitive Evaluation Work

Faculty members repeatedly perform the same tasks:

- Read answers
- Compare with rubric

- Assign marks
- Write feedback

This repetition causes:

- Fatigue
 - Reduced consistency
 - Human error
 - Slower evaluation
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3.4 Manual Grade Uploading

After evaluation, faculty still need to:

- Open Canvas gradebook
- Enter marks manually
- Upload comments individually

This is highly time-consuming.

3.5 Delayed Student Feedback

Because manual grading takes time:

- Students receive delayed feedback
 - Learning improvement slows down
 - Course progression becomes inefficient
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4. Solution Provided by This System

The AI Evaluator system solves these issues using automation.

Instead of manually handling assignments, the system automates the entire workflow.

The faculty member only needs to:

1. Configure API keys
2. Run two Python files

The system handles everything else automatically.

5. Main Objectives of the System

The primary objectives are:

5.1 Reduce Faculty Workload

The system minimizes repetitive administrative work.

Faculty can focus more on:

- Teaching
- Mentoring
- Course improvement
- Student interaction

instead of repetitive grading tasks.

5.2 Increase Evaluation Speed

The AI can evaluate assignments much faster than manual workflows.

Large batches of submissions can be processed efficiently.

5.3 Maintain Consistency

Human evaluation often changes due to:

- Fatigue
- Mood
- Time pressure

AI-based rubric evaluation improves consistency.

5.4 Improve Feedback Quality

The system generates:

- Overall feedback
- Strengths
- Areas for improvement

This helps students understand their mistakes better.

5.5 Automate Administrative Tasks

Tasks such as:

- Downloading
- Organizing
- CSV creation
- Grade uploads

are fully automated.

6. Core Concept of the System

The system works like a virtual teaching assistant.

It performs the repetitive operational tasks while the faculty supervises the academic process.

The AI does not randomly assign marks.

Instead, it:

- Reads assignment instructions
- Reads rubric criteria
- Reads student submissions
- Evaluates according to rubric logic
- Generates structured grading

This creates a more scalable academic workflow.

7. High-Level Workflow

The entire workflow can be visualized as:

Canvas LMS



Fetch Assignment Details



Download Student Submissions



Organize Student Files



Read Documents and Code Files



AI-Based Rubric Evaluation



Generate Scores and Feedback



Upload Grades to Canvas



Create CSV Reports

8. Major Components of the System

The project consists of two main programs.

8.1 Assignment Downloader

File:

Assignment-Downloader.py

Purpose

This module automatically downloads all student submissions from Canvas.

Main Functions

Connects to Canvas LMS

The downloader uses:

- Canvas API token
- Course ID
- Assignment ID

to connect directly to Canvas.

Fetches All Submissions

It automatically retrieves:

- Student names

- Uploaded files
 - Text submissions
 - URL submissions
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Downloads Files in Parallel

Unlike manual downloading, the system downloads multiple files simultaneously.

Advantages:

- Faster downloads
 - Reduced waiting time
 - Better performance for large classes
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Organizes Student Data

The system automatically:

- Names files correctly
- Organizes content
- Creates structured folders

No manual sorting required.

Creates Backup ZIP

All submissions are automatically compressed into:

submissions.zip

This acts as a backup archive.

Extracts Files Automatically

Files are extracted into:

submissions/

ready for evaluation.

8.2 AI Evaluator

File:

Purpose

This module performs intelligent evaluation using AI.

Main Functions

Reads Assignment Details

The evaluator automatically fetches:

- Assignment question
- Assignment description
- Rubric criteria
- Maximum marks

from Canvas.

Reads Student Submissions

The system supports many file types.

Examples:

Type	Supported
PDF	✓
DOCX	✓
TXT	✓
Python	✓
Java	✓
C/C++	✓
HTML/CSS	✓
SQL	✓

AI-Based Evaluation

The evaluator uses:

Groq

to analyze student work.

The AI evaluates:

- Correctness
 - Completeness
 - Clarity
 - Creativity
 - Rubric alignment
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Continuous Scoring

Traditional grading often uses fixed grades.

Example:

Excellent

Good

Average

This system uses detailed continuous scoring.

Example:

83/100

74/100

91/100

This provides finer academic distinction.

Generates Feedback Automatically

Each student receives:

- Overall feedback
- Strengths
- Improvement suggestions

This reduces repetitive faculty work.

Uploads Grades Automatically

The system directly uploads:

- Marks
- Rubric scores
- Feedback comments

back into Canvas.

No manual grade entry required.

Creates CSV Reports

A detailed CSV report is generated containing:

- Student names
- Marks
- Grades
- Feedback
- Strengths
- Improvements

Useful for:

- Department records
 - Accreditation
 - Analytics
 - Faculty documentation
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9. Advantages Over Traditional Evaluation

9.1 Time Efficiency

Traditional grading:

- Several hours or days

AI-assisted grading:

- Much faster workflow

This becomes extremely useful for:

- Large classrooms
 - Multiple sections
 - Frequent assignments
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9.2 Reduced Manual Work

Manual operations eliminated:

- Downloading files individually
 - Organizing folders
 - Writing repetitive comments
 - Uploading grades manually
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9.3 Better Scalability

The system handles large batches efficiently.

Example:

Students Manual Difficulty AI System

20	Easy	Easy
100	Difficult	Efficient
500	Extremely difficult	Manageable

9.4 Consistent Evaluation

Human grading often changes because of:

- Tiredness
- Time pressure
- Subjective variation

AI rubric evaluation improves grading consistency.

9.5 Faster Student Feedback

Students receive:

- Marks quickly
- Feedback faster

- Better learning support

This improves academic engagement.

9.6 Better Academic Documentation

The CSV reports provide structured academic records useful for:

- Internal audits
 - Accreditation
 - Department analytics
 - Performance tracking
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9.7 Supports Multiple Submission Types

Traditional workflows struggle with mixed file types.

This system handles:

- Documents
- Source code
- Text submissions
- URLs

within one workflow.

10. Educational Impact

This system supports modern digital education.

It is especially valuable for:

- Online learning
- Hybrid learning
- Large universities
- Technical courses
- Programming education

It enables institutions to scale evaluation workflows more effectively.

11. Practical Use Cases

Universities

Managing large semester assignments.

Programming Courses

Evaluating source code submissions.

MOOCs and Online Learning

Handling large student volumes efficiently.

Internal Assessments

Automating rubric-based grading.

Faculty Research and Analytics

Using generated reports for academic insights.

12. Platform Independence

The system works on:

Operating System Supported

Windows 

Ubuntu/Linux 

macOS 

Because the system is written in:

Python

it behaves consistently across platforms.

13. Security and Privacy

The system uses:

- Canvas API tokens

- Groq API keys

These credentials allow secure communication with Canvas and AI services.

Users should:

- Keep keys private
 - Avoid public sharing
 - Remove keys before publishing code
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14. Limitations

Although powerful, the system has limitations.

AI Cannot Fully Replace Faculty

Faculty supervision is still recommended.

Image Understanding is Limited

Image-based submissions are only partially supported.

Internet Dependency

The system requires internet connectivity.

15. Final Conclusion

The Canvas Assignment AI Evaluator System is a modern academic automation framework designed to simplify assignment management and evaluation.

It combines:

- Canvas LMS integration
- Automated downloading
- AI-based evaluation
- Rubric grading
- Feedback generation
- Grade uploading
- Report generation

into one integrated solution.

The system significantly reduces repetitive faculty workload while improving evaluation speed, consistency, and scalability.

Rather than replacing educators, the platform acts as an intelligent academic assistant that helps faculty manage large digital classrooms more efficiently.